2FiA: Towards WiFi Sensing-Based Authentication with Unique Biometrics

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Presented by Matthew Chen

Remember This?

The heartbeat sensor from Call of Duty.







Image Credits:

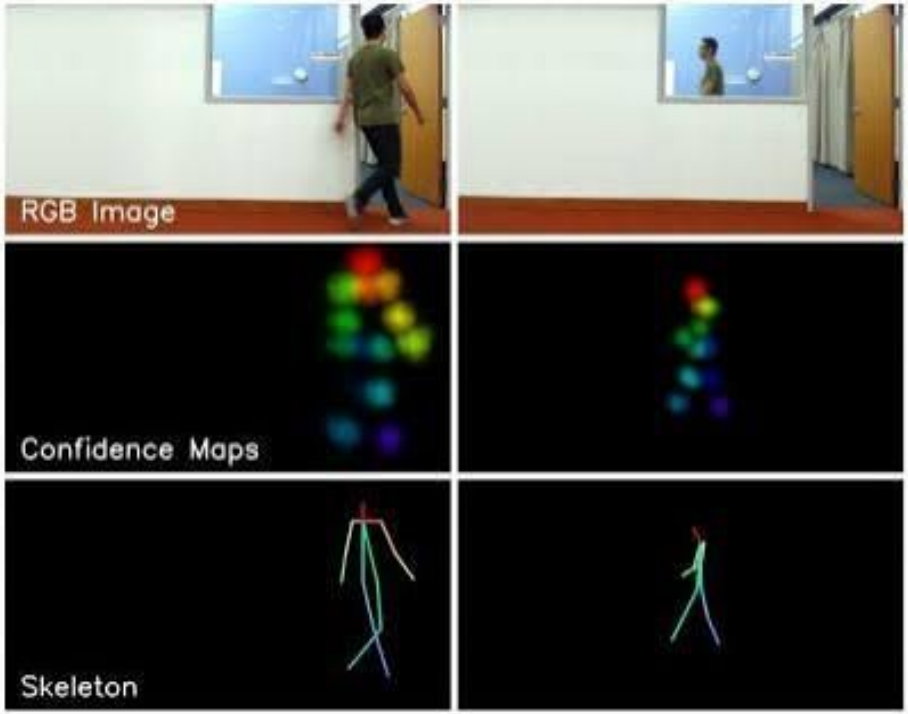
Reddit User (u/dineshsura). (2022). My suggestions about Heartbeat Sensor (SnD) [Online image / forum post]. Reddit. https://www.reddit.com/r/CallOfDutyMobile/comments/wj5c3n/my_suggestions_about_heartbeat_sensor_snd/

Call of Duty Wiki. (n.d.). *Heartbeat Sensor*. Fandom. https://callofduty.fandom.com/wiki/Heartbeat_Sensor

Or This

See Through Walls With Wifi!

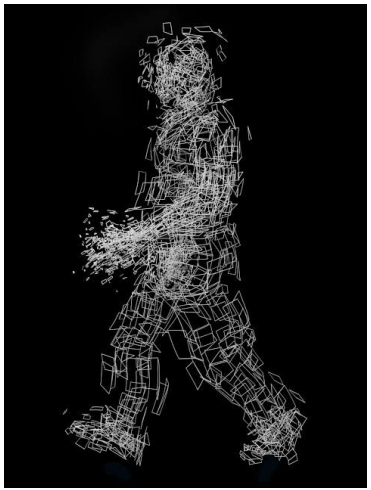
https://people.csail.mit.edu/fadel/papers/wivi-paper.pdf



2FiA - 2 factor Wi-Fi sensing based authentication

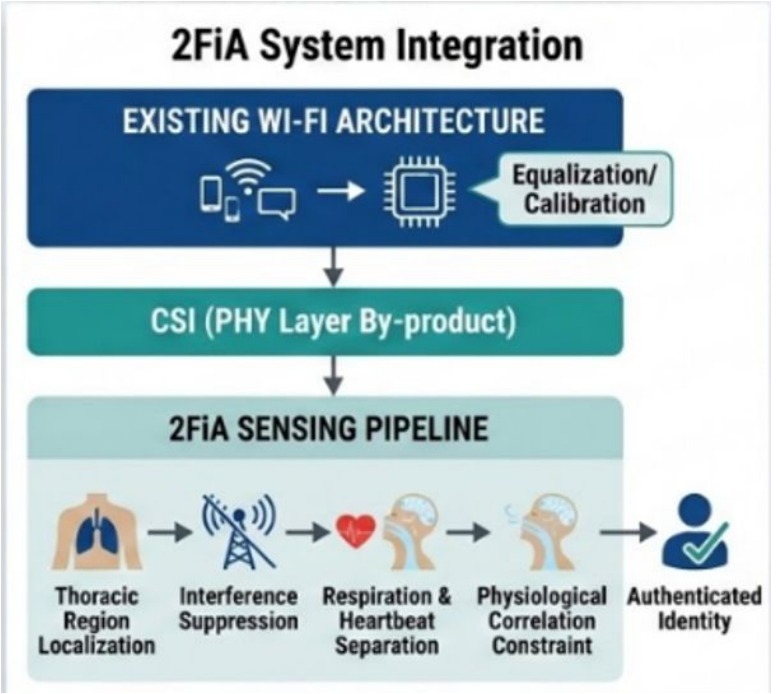
With Wi-Fi CSI, the channel measurements your chip already exposes, can be used for more than sensing. 2FiA explores the first dual biometrics Wi-Fi sensing based authentication system. It uses metrics like your breathing patterns (primary) and involuntary heart beat pattern (secondary) to authenticate and identify users.





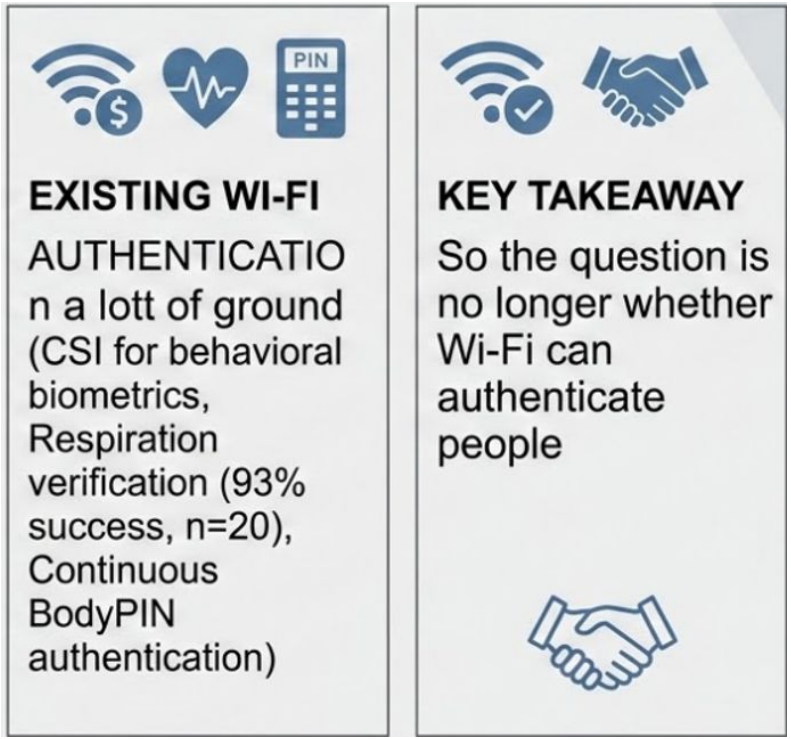
Where it fits in architecture

- Wifi was built for communication
  - CSI is the PHY layer's channel estimate, a by product used for equalization and calibration
- The paper treats that by product as a sensor
  - Reuses infrastructure/components meant for one job for another
- How it works:
  - Chip exposes CSI
  - Sensing pipeline: localizes thoracic region, suppresses interference from non thoracic areas, separates respiration from co-located weaker heartbeat component, uses physiological correlation constraint for validating heartbeat



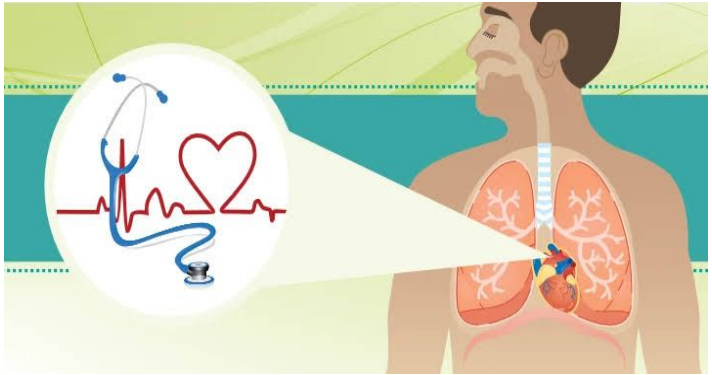
Why does this matter?

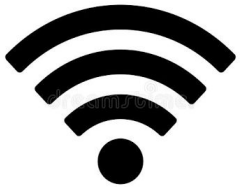
- Passwords, tokens, and other biometrics all have costs
  - E.g. user effort, extra hardware, privacy, etc.
- Earlier Wi-Fi authentication covered a lot of ground
  - One line uses CSI to capture behavioral biometrics without sensors or wearables
  - Another uses respiration for continuous verification (93% success with n=20 subjects)
  - A third line targets continuous authentication (BodyPIN)
- So the question is no longer whether Wi-Fi can authenticate people



The gap that motivates 2FiA

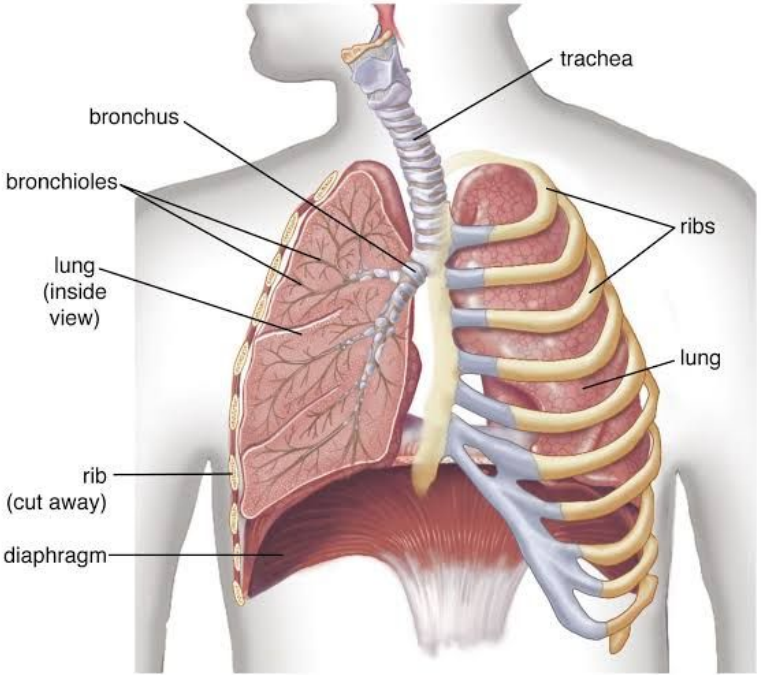
- How hard is the biometric to imitate?
  - Breathing patterns are easy to observe and mimic
- A recent SoK on Wi-Fi CSI biometrics (arXiv 2511.11381) calls for more rigorous security evaluation of this whole area
  - That places 2FiA on the field
- Heartbeat is much harder to imitate, but its signal is far weaker
  - Which is the core story of 2FiA





Key Idea

- Add heartbeat as a second factor, giving you “2 factors in 1 signal”
- The hard part is getting a usable heartbeat metric out of CSI
    - The pipeline localizes the thoracic region, suppresses co-located interference, separates respiration from the weaker heartbeat, and uses a physiological correlation constraint to validate heartbeat signals



What it reveals about the research landscape

- Security vs Sensing tradeoff
  - The stronger the biometric, the harder it is to sense
- Sensing as an attack surface
  - If Wi-Fi can read your heartbeat, an attacker can potentially read/reply it too
- Cross-layer reuse
  - The same pattern is used in localization, gesture recognition, and pose estimation
  - CSI is a general purpose sensor
- Evaluation rigor
  - The SoK shows the community is still working out threat models and metrics

Possible research opportunities

- Robustness across environments
  - People in motion
  - Multiple people in a room
- Adversarial and replay attacks against CSI
- Privacy
- Continuous authentication and multi factor combinations with other wireless modalities